

Research paper

Hybrid dual-head IDS: A lightweight and explainable deep learning DDOS/DOS framework for zero-day and imbalanced attack detection in IoT networks

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ABSTRACT

The rapid expansion of IoT has escalated security risks of DDoS and DoS for cyber-physical systems, but deep learning-based Intrusion Detection Systems (IDS) face challenges like poor handling of imbalanced datasets, inability to detect zero-day attacks, lack of transparency and scalability. This paper proposes Hybrid Dual-Head IDS, a lightweight, explainable deep learning framework to address these issues. The framework employs a dataset-adaptive Wasserstein GAN with Gradient Penalty per class (auto-selecting jitter, bootstrapping, or WGAN-GP as determined by sample size (N_{eff})-scaled Kolmogorov–Smirnov test) and a functionally partitioned dual-head architecture with a hybrid loss function. A ZDS Ratio that is zero-day separation for vanishing FPR and per-class SHAP values (that can be linked to the dual-head thresholds) is used by the operational analysts. Evaluated on CIC-IoT2023, BoT-IoT, and ToN-IoT datasets, Hybrid Dual-Head IDS achieves over 99.5% accuracy and F1-scores, perfect scores on the imbalanced BoT-IoT dataset, and high zero-day detection capability. The evaluation is further strengthened by introducing a true zero-day holdout protocol, a component-wise ablation study, five-seed statistical validation, direct synthetic-data fidelity analysis, and ROC-AUC and threshold-sensitivity evaluation. In true zero-day holdout setting, the model achieved ZDS ratios of 175,900.50x/1094.12x on CIC-IoT2023 (binary/multiclass), 7260,770.86x/12791298.51x on BoT-IoT, and 223.95x on ToN-IoT. Results further show stable behavior across random seeds, high-quality synthetic augmentation where needed, and very low false positive rates under ROC-based threshold selection. The compact size (≈ 0.5 MB) enables efficient network protection through high throughput which provides an effective practical solution for IoT network security against known and unknown threats.

1. Introduction

The Internet of Things (IoT) has revolutionized digital connectivity by merging billions of smart devices into our daily lives. The smart devices in homes and wearable technology and industrial control systems (IIoT) and critical healthcare infrastructure produce vast amounts of data which results in better operational performance and innovative technological developments. The extensive network of connected devices has created a complex system which hackers can target through multiple entry points [1]. The Internet of Things (IoT) devices remain

vulnerable to cybercriminals because they lack strong security protocols and have limited processing power and memory capacity and receive infrequent software updates. The limited processing power and memory capacity of IoT devices combined with their weak security measures and infrequent software updates makes them highly susceptible to cyber-attacks. Therefore, the defense of IoT networks against modern and evolving cyber threats has become an essential cybersecurity problem for current times [2]. Intrusion Detection Systems (IDS) is a fundamental security measure which monitors network traffic to find out abnormalities and alert potential dangers. Conventionally, IDS

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systems exist in two main categories which include signature-based and anomaly-based systems. Signature-based systems function well for known attack detection because they compare network traffic to a pre-existing database of threat signatures. Their primary drawback, however, is their inability to detect novel or zero-day attacks for which no signature exists. In contrast, anomaly-based systems create a baseline of typical network activity and alert on any deviations that may indicate potential intrusions. Although this system can potentially identify zero-day threats by detecting deviations from normal behavior, they frequently generate high false-positive rates. Therefore system becomes overwhelmed which causes security analysts to face excessive workloads and damages the trust in system reliability.

In recent years, machine learning (ML) and deep learning (DL) have transformed the field of intrusion detection through their implementation which provides advanced security capabilities beyond conventional detection systems [3]. Research has shown that CNNs can extract spatial features from network traffic while RNNs can extract temporal patterns according to [4]. Research studies show that hybrid models represent the key solution which organizations need to boost their performance outcomes. The models which include structural hybrids that use CNN-LSTM frameworks for spatial and sequential analysis or to more complex architectural designs [5]. IDS research now uses advanced designs that let internal operations perform tasks by evaluating feature importance through attention mechanisms and handling multiple objectives through multi-task architectures. Privacy-preserving IDS frameworks for secure data sharing and decentralized sensing in IoT network are explored to be adopted [6]. The current IDS research integrates these modern approaches with multiple privacy protection techniques. The implementation of DL-based IDSs faces three essential challenges which reduce their effectiveness in real-world applications. The first major issue arises from the severe class imbalance problem. In any real-world network, benign traffic vastly outnumbers malicious traffic, and within the malicious category, some attack types are exceedingly rare. The training process becomes biased because of this imbalance which results in poor performance of models on minority attack classes. Findings in [7] suggest that classification performance is significantly affected by interactions of resampling and feature selection methods. The traditional oversampling method SMOTE exists but Generative Adversarial Networks (GANs) have proven to be a better solution for creating realistic synthetic data to balance complex datasets [8]. Second, The detection of zero-day attacks continues to be an unresolved challenge in the field of reliable attack detection. The training data of most supervised models restricts their ability to recognize new threat patterns that were not present in their training data. This requires a hybrid method which combines supervised classification precision with the novelty-detection capabilities of unsupervised methods that use autoencoders or specific anomaly measurement methods [9]. Third, The black box nature of Deep learning models creates transparency problems which results in trust issues. Security analysts need to understand the reasoning behind IDS identification of particular malicious activities because IDS effectiveness depends on their ability to do so. Therefore, the use of XAI techniques such as SHAP (SHapley Additive exPlanations) has become increasingly important because these methods provide insights to understand decisions of a model [10]. The same set of principles are being applied to design secure, resilient and trustworthy Internet of Things systems [11].

To address these challenges, the Hybrid Dual-Head IDS framework is proposed as a lightweight deep learning system for IoT network intrusion detection with explainable outputs. The framework simultaneously tackles class imbalance, zero-day detection, and model transparency. WGAN-GP is employed to improve minority class samples through robust representations of rare attacks. For each class, it auto-selects the most appropriate generation feature space, including Gaussian jitter to exploit boundary features, bootstrapping for integer-valued feature spaces representing protocols, full WGAN-GP for large, smooth distributions, etc. with corresponding (N_{eff})-scaled KS-statistics thresholds,

ensuring that the generated samples always are valid. A dual-head architecture combines supervised classification via hybrid loss function for precise multiclass threat identification with unsupervised anomaly detection for zero-day attack identification. The Zero-Day Separation (ZDS) Ratio is introduced as a novel metric for evaluating zero-day attack detection effectiveness. The ZDS Ratio is derived from hypothesis testing. Complete interpretability is achieved through SHAP, revealing attack-class-dependent features influencing predictions. A strict true zero-day holdout protocol is implemented, removing one attack class entirely from model development for final evaluation only. Additionally, a component-level ablation study is conducted, all experiments are run with multiple seeds for statistical strength, synthetic data fidelity is verified, and ROC-AUC with threshold sensitivity analyses are performed on false positives.

The research delivers following main contributions to the field:

1. **A Novel Hybrid Dual-Head IDS Framework:** The proposed lightweight dual-head deep learning system combines supervised classification with unsupervised anomaly detection to provide complete threat protection.
2. **Advanced Data Augmentation for Imbalance:** The WGAN-GP proves effective for class imbalance reduction through its ability to create realistic synthetic minority attack class samples which enhance detection performance. Perform feature-wise Wasserstein distance analysis, KS-test and t-SNE to further validate quality of synthetic data.
3. **A New Metric for Zero-Day Detection:** We introduce the Zero-Day Separation (ZDS) Ratio, a novel metric to quantitatively evaluate a model's ability to distinguish unseen attacks from benign traffic. Including under the true holdout setting.
4. **Rigorously Controlled Empirical Validation:** All experiments are repeated with 5 random seeds with mean $\pm$ SD / median $\pm$ IQR. An ablation study of the 5 different configurations to validate the individual contributions of the different components. A direct assessment of the synthetic data's fidelity (rather than just downstream accuracy) validates augmentation quality. ROC-AUC and threshold-sensitivity analysis provides operational deployability. True zero day holdout protocol with quantification of the ZDS separation to provide reproducible novelty detection benchmarks.

The novelty of this work is a unified, lightweight, and explainable dual-head intrusion detection framework that jointly handles imbalanced DDoS/DoS classification and zero-day anomaly detection in IoT networks. Unlike most studies that concentrate solely on supervised classification accuracy or GAN-based data balancing, the proposed framework combines WGAN-GP-driven adaptive augmentation, a supervised classification head, a parallel anomaly-detection head, SHAP-based explainability, and deployment-oriented efficiency analysis in a single pipeline. The study further introduces the Zero-Day Separation Ratio as an additional score-space metric for evaluating unseen threats and validates zero-day capability through a true attack-class holdout protocol. The framework is supported by ablation analysis, five-seed reproducibility, direct assessment of synthetic-data fidelity, and ROC-AUC/FPR threshold analysis across the CIC-IoT2023, BoT-IoT, and ToN-IoT datasets.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 provides a detailed description of the proposed methodology, including its architecture and the ZDS metric. Section 4 presents the experimental setup, a thorough analysis of the results. Section 5 present the discussion and Finally, Section 6 concludes the paper and outlines potential directions for future research.

2. Related work

The field of intrusion detection systems (IDS) has undergone significant evolution, driven by the integration of machine learning (ML) and

deep learning (DL) techniques to counter increasingly sophisticated cyber threats, including Denial-of-Service (DoS), Distributed Denial-of-Service (DDoS), and zero-day attacks. Traditional IDS methods based on signature and anomaly detection struggle to detect new threats and generate excessive false positives while facing scalability issues when deployed in Internet of Things (IoT) networks with their dynamic nature [12]. The growing number of IoT devices has made these problems worse because it creates diverse network traffic patterns and produces extreme class differences between normal and malicious data points [13]. Recent studies on interpretable lightweight deep learning in industrial and cyber-physical systems including a CNN-GRU-based approach for intrusion detection in industrial IoT scenarios have inspired this work [14]. In consideration, GANs have become a revolutionary technology which produces synthetic data for dataset expansion and adversarial training to enhance robustness and address data scarcity and unbalanced datasets [15]. The section evaluates essential developments in GAN-based IDS through three main trends which include data augmentation methods for imbalance correction and robustness improvements from adversarial and unsupervised learning and the creation of sophisticated hybrid systems while identifying ongoing challenges in unification and explainability and real-time adaptation and scalability.

2.1. GANs for data augmentation and imbalance mitigation

A core application of GANs in IDS is addressing class imbalance, a pervasive issue in network datasets where rare attack samples lead to biased models with poor minority-class detection. Standard classifiers usually perform better on majority classes, so recent reviews recommend using data-level, algorithmic, or hybrid methods to address weak minority-class detection [16]. GANs learn underlying data distributions to produce realistic synthetic samples, augmenting minority classes and enabling balanced training without relying solely on costly, privacy-sensitive real-world data [17]. Early efforts demonstrated the efficacy of training models exclusively on GAN-generated data, achieving high accuracy on real test sets [18]. Specialized variants have proven particularly effective for tabular network flow data. For instance, the Conditional Tabular GAN (CTGAN) conditions generation on class labels to create balanced datasets, as seen in Alabsi et al. (2023), where CTGAN augments IoT datasets like BOT-IOT for binary DoS/DDoS detection, improving accuracy through staged preprocessing, generation, and ML/DL classification [19]. Similarly, S. Rahman et al. (2024) proposed SYN-GAN, a GAN trained on p datasets (UNSWNB15, NSL-KDD and BOT-IOT), yielding up to, 90, 84 and 100% accuracy on for threats including DoS, DDoS, and reconnaissance [18]. Wasserstein GANs (WGANs), valued for training stability, have been applied to oversample minority classes in CICIDS2017 [20] and The Conditional Tabular GAN (CTGAN) augment datasets like BoT-IoT and NF-v2-Bot-IoT for detecting rare attacks such as DoS, DDoS, and theft [19,21]. Hybrid datasets combining original and synthetic samples are common, extending to diverse contexts like MQTT protocols [22] and non-network data via Mel spectrogram transformations [23]. Refinements include combining CTGAN oversampling with undersampling techniques like Edited Nearest Neighbors (ENN) for cleaner datasets [24]. While these approaches enhance generalization and ethical data handling, they often remain offline, dataset-specific, and overlook multiclass scenarios or real-time needs. Hybrid resampling methods are expected to improve minority-class detection while still preserving computational efficiency [25]. A key research challenge is the use of uniform generators and accuracy-based assessment metrics in current augmentation approaches, which are unable to reflect the fidelity of synthetic network traffic features, the statistical validity of the data, and true generalization of synthetic network traffic data. This research overcomes these issues through dataset-adaptive augmentation selection with explicit per-feature statistical gates, direct fidelity validation, calibrated thresholds, and by generating synthetic samples that

maintain the empirical structure of all continuous, integer-valued and mode-concentrated features.

2.2. Enhancing IDS robustness with adversarial and unsupervised learning

GANs also fortify IDS against evasive and novel threats through adversarial training and unsupervised anomaly detection. Adversarial training uses GANs to craft deceptive samples that challenge the detector, refining decision boundaries for resilience. The GADoT system employs a Generative Adversarial Network (GAN) to produce adversarial DDoS samples achieves high accuracy on standard datasets yet fails to detect >60% of malicious network flows when attacked by adversarial samples [26]. Research studies have used this method by adding adversarial examples produced through Fast Gradient Sign Method (FGSM) to their training process to enhance model resistance against standard and adversarial machine learning (AML) attacks [27,22,28]. Research using GANs operates in two parallel directions that include both unsupervised and semi-supervised applications. The GAN receives training only from normal traffic data in this approach. The discriminator model or autoencoder receives training to identify complex patterns in normal traffic data. The system identifies traffic anomalies through its ability to detect major deviations from its learned baseline traffic patterns. The method proves effective for zero-day attack detection since it operates without needing any pre-existing attack signature information. GAN-AE represents a notable example because it employs an adversarially trained autoencoder to identify MQTT network anomalies through reconstruction error measurement [29], and other systems that use GAN discriminator scores to detect abnormal network flows in IoT and SDN environments with high noise levels [30,31]. There are significant research gaps in current zero-day IDS evaluations, which rely on incomplete and often qualitative evaluation settings that do not withhold all unseen attack classes, and therefore cannot quantitatively and reliably measure the amount of zero-day attack detection separation. Our proposed model synthesizes these concepts into a hybrid framework. While its primary training is supervised on a GAN-augmented dataset, it incorporates a dedicated anomaly detection head and a quantified zero-day separation metric. This dual architecture allows our IDS to accurately classify known attack types while simultaneously scoring all traffic for anomalous characteristics, thereby combining the precision of supervised learning with the quantified novel threat detection capabilities of unsupervised methods.

2.3. Advanced hybrid architectures and emerging trends

To tackle complex threats, recent IDS integrate GANs into hybrid architectures capturing spatial-temporal dependencies, alongside emerging focuses on explainability, online learning, and scalability. The combination of GANs with DL components such as CNN-LSTM/GRU [22, 32] and Transformer-GAN [33] and stacked ensembles [34,35] produces better results in hybrid models. However, there remain significant challenges to IoT IDS including heavy dependence on large amounts of labeled training data, poor adaptability to new attacks, and lack of interpretability [36]. The black-box nature of GAN-enhanced models becomes transparent through Explainable AI (XAI) methods which include SHAP and LIME to identify essential features [37,24], as Security analysts can validate and understand the model's decision-making process through this approach. Also SHAP-enabled explainable deep learning models for Industry 5.0 CPS security [38]. Despite progress, gaps include specialization to binary or multiclass tasks without unification [23], underexplored real-time adaptation [39], and inefficiencies in large-scale deployments [40]. Most rely on static training, limiting dynamic IoT applicability. A major research gap is that current GAN-based IDS approaches are designed for narrow tasks, so binary and multiclass detection often require separate models rather than one practical unified framework. Our Unified Successful GAN-IDS bridges these by offering a cohesive WGAN-GP-integrated hybrid model

supporting binary and multiclass classification on different datasets. Innovations include SHAP for explainability, simple online learning for real-time adaptation, and batch processing for scalability. Achieving High accuracy, high minority F1-scores, and robust zero-day handling, it outperforms priors in robustness, interpretability, and practicality.

Although earlier research has examined GAN-based data augmentation, deep learning classifiers, anomaly detection, and explainability for intrusion detection systems, these areas are often studied independently. Many existing approaches emphasize classification accuracy, while overlooking true zero-day attack scenarios in which entire attack classes are held out during testing. In addition, synthetic data is usually evaluated only through its effect on classifier performance rather than through direct fidelity assessment. Moreover, few studies that combine a zero-day evaluation, a quality assessment of synthetic data, a statement on statistical reproducibility, a threshold-based analysis of false positives and some considerations for a light-weight deployment. The proposed Hybrid Dual-Head IDS aims to address these limitations by combining adaptive WGAN-GP augmentation, dual-head classification and anomaly detection, SHAP-based explainability, true zero-day holdout testing, and efficiency analysis for deployment across three modern IoT datasets.

3. Proposed methodology: the hybrid dual-head IDS framework

The proposed framework in Fig. 1, named Hybrid Dual-Head IDS, is a multi-stage deep learning pipeline designed for robust and explainable intrusion detection in modern network environments. The system is engineered to address two of the most critical challenges in the field: severe class imbalance and the detection of zero-day threats. Its efficacy and generalizability are validated across three diverse and challenging datasets: CIC-IoT2023, BoT-IoT, and ToN-IoT. The framework architecture consists of five stages which include (1) Adaptive Data Pre-processing followed by (2) WGAN-GP-based Data Augmentation and then (3) Hybrid Dual-Head Deep Learning Detection and (4) Hybrid Training, and finally (5) Comprehensive Evaluation and Explainability. This design includes modular stages which operate efficiently for both research applications and real-world implementation.

3.1. Datasets and adaptive preprocessing

The evaluation framework uses three separate datasets which assess its performance through different testing scenarios. The CIC-IoT2023 dataset delivers a modern complex testing environment through its 42 features which detects multiple attack types including different DDoS and normal attacks. The BoT-IoT dataset contains 38 features which show severe class imbalance while including different DDOS, DoS and

Normal attack classes. Lastly, The ToN-IoT dataset presents a diverse testing environment through its 41 attributes which stem from a big IoT/IIoT testbed. The quality analysis in our adaptive preprocessing stage first checks for skewness in the data. skewness is computed according to the formula in Eq. (1):

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^{3/2}} \quad (1)$$

where x_i are data points, $\bar{x}$ is the sample mean, and n is the sample size.

This measures the dataset's asymmetry (positive for right skew, negative for left skew, zero for symmetric).

Skewness removal takes place through Quantile Uniform Transformation. The quantile uniform transformation transforms a random variable X (your dataset) with a cumulative distribution function (CDF) $F_X(X)$ into a uniform random variable U on $[0,1]$ represent in Eq. (2):

$$U = F_X(X) \quad (2)$$

where $F_X(X) = P(X \leq x)$ is the CDF of X , and $U \sim \text{Uniform}[0,1]$.

This is followed by label encoding and a stratified split of the full dataset D into training D_{train} , validation D_{val} , and testing D_{test} subsets, such that $D = D_{\text{train}} \cup D_{\text{val}} \cup D_{\text{test}}$, with proportions of 60%, 20%, and 20%, respectively. For instance, the BoT-IoT dataset of 802,951 samples was partitioned into 481,770 training, 160,590 validation, and 160,591 testing samples.

To avoid data leakage and to avoid over-estimating model performance, all data-dependent preprocessing was fit on the training data only. The validation and test partitions were kept completely unseen during model training, augmentation, and hyperparameter selection. Specifically, the quantile transformation and scaling parameters were estimated from D_{train} only and then applied unchanged to D_{val} and D_{test} . WGAN-GP was also trained only on minority-class samples from D_{train} , and synthetic samples were added only to the training set. No synthetic sample was included in the validation or test set. This split-before-augmentation protocol prevents information from validation or test data from influencing the learned representation. A stricter protocol is introduced for evaluating zero-day detection. Rather than computing anomaly separation only among classes already represented during training, one complete attack class is removed from model development and reserved exclusively for final evaluation. This produced a true holdout-based zero-day setting in which the anomaly head was tested on genuinely unseen attack traffic. For CIC-IoT2023, the withheld attack class was DDoS-SYN_Flood. For BoT-IoT, the withheld class was DDH. For ToN-IoT, the withheld class was ddos. After this removal, the

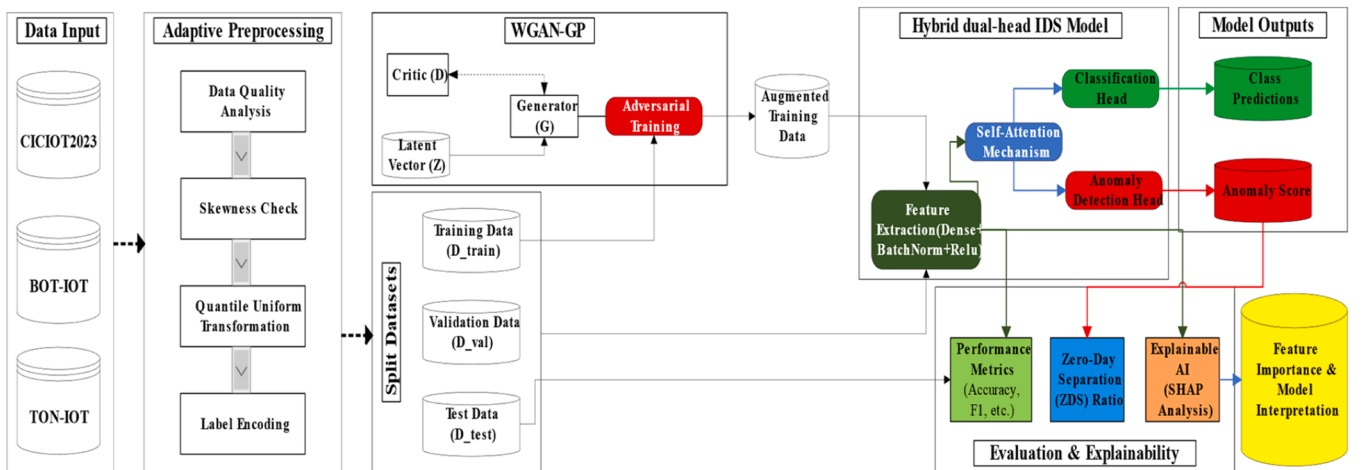


Fig. 1. Block diagram of hybrid dual-head IDS.

remaining data were processed and split as described above. The withheld attack class was then reintroduced only during the final anomaly-based zero-day evaluation. Table 1 summarizes the comprehensive performance metrics across all datasets and tasks, including the experimental zero-day protocol results.

3.1.1. Strict separation between training and evaluation pipelines

To address the risk of leakage in synthetic-data augmentation, the proposed framework uses two strictly separated pipelines: a training pipeline and an evaluation pipeline. The training pipeline includes data-dependent preprocessing, WGAN-GP training, synthetic sample generation, classifier training, anomaly-head optimization, and validation-based hyperparameter or threshold selection. The evaluation pipeline uses only the frozen preprocessing parameters, the trained model, and the selected decision threshold. No WGAN-generated sample is included in the validation or test partitions, and no validation or test sample is used to train the WGAN-GP model. Algorithm 1 explains the complete process which clearly mentions the separation.

3.2. Data augmentation with Wasserstein GAN with gradient penalty (WGAN-GP)

To combat class imbalance, we employ a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP). This model consists of a generator, G , which maps a random latent vector z drawn from a prior distribution $p(z)$ (e.g., $z \sim N(0, I)$) to the data space, $\tilde{x} = G(z)$, and a critic, D , which outputs a scalar score indicating the realness of a sample. The WGAN-GP was chosen for its training stability. The critic's loss function minimizes the Wasserstein-1 distance with a gradient penalty to enforce the 1-Lipschitz constraint shown in Eq. (3):

$$L_{\text{Critic}} = E_{\tilde{x} \sim P_g}[D(\tilde{x})] - E_{x \sim P_r}[D(x)] + \lambda E_{\tilde{x} \sim P_g} \left[\left(\|\nabla_{\tilde{x}} D(\tilde{x})\|_2 - 1 \right)^2 \right] \quad (3)$$

Here, P_r is the real data distribution, P_g is the generator's distribution, and $\tilde{x}$ is an interpolated sample. The generator's objective is to produce samples that the critic scores as real, guided by the loss shown in following Eq. (4):

$$L_{\text{Generator}} = -E_{\tilde{x} \sim P_g}[D(\tilde{x})] \quad (4)$$

The WGAN-GP is trained for 50 epochs, with convergence monitored via losses (e.g., Gen: -2.2021, Disc: 1.4607 on BoT-IoT). This process synthesizes high-quality samples for minority classes, such as augmenting the 'Normal' class in the BoT-IoT binary task from 286 to 858 samples.

The role of augmentation was not assessed only through downstream classification accuracy; direct synthetic-data fidelity validation is additionally performed. Specifically, the generated minority-class samples were compared against real data using feature-wise Wasserstein distance, two-sample Kolmogorov–Smirnov testing, and two-dimensional t-SNE visualization. This additional validation was introduced to ensure that the synthetic samples were statistically meaningful rather than simply useful in a narrow predictive sense.

Algorithm 1

Leakage-controlled training and evaluation protocol.

Require: Real dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, minority classes $\mathcal{C}_{\min}$, train/validation/test split ratio $(\rho_{\text{train}}, \rho_{\text{val}}, \rho_{\text{test}})$
Ensure: Trained IDS model f_θ , final test metrics $\mathcal{M}$

- 1 $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}) \leftarrow \text{StratifiedSplit}(\mathcal{D}, \rho_{\text{train}}, \rho_{\text{val}}, \rho_{\text{test}})$
- 2 $\theta_{\text{prep}} \leftarrow \text{FitPreprocessing}(\mathcal{D}_{\text{train}})$
- 3 $\tilde{\mathcal{D}}_{\text{train}} \leftarrow \theta_{\text{prep}}(\mathcal{D}_{\text{train}})$, $\tilde{\mathcal{D}}_{\text{val}} \leftarrow \theta_{\text{prep}}(\mathcal{D}_{\text{val}})$, $\tilde{\mathcal{D}}_{\text{test}} \leftarrow \theta_{\text{prep}}(\mathcal{D}_{\text{test}})$
- 4 $\mathcal{D}_{\text{train}}^{\min} \leftarrow \{(x, y) \in \mathcal{D}_{\text{train}} : y \in \mathcal{C}_{\min}\}$
- 5 $G \leftarrow \text{TrainWGAN} - \text{GP}(\mathcal{D}_{\text{train}}^{\min})$
- 6 $\mathcal{S}_{\min} \leftarrow G(z), z \sim p(z)$
- 7 $\tilde{\mathcal{D}}_{\text{train}}^{\text{aug}} \leftarrow \tilde{\mathcal{D}}_{\text{train}} \cup \mathcal{S}_{\min}$
- 8 $f_\theta \leftarrow \text{TrainIDS}(\tilde{\mathcal{D}}_{\text{train}}^{\text{aug}})$
- 9 $\tau^* \leftarrow \text{TuneThresholds}(f_\theta, \tilde{\mathcal{D}}_{\text{val}})$
- 10 **Freeze** $\theta_{\text{prep}}, G, f_\theta, \tau^*$
- 11 $\mathcal{M} \leftarrow \text{Evaluate}(f_\theta, \tilde{\mathcal{D}}_{\text{test}})$
- 12 **return** $f_\theta, \mathcal{M}$

3.3. The hybrid dual-head detection model

The core of our framework is the Hybrid Dual-Head IDS model, a deep neural network designed to process an input feature vector $X \in \mathbb{R}^d$. The architecture begins with a deep Feature Extractor that learns a foundational representation, $F_{\text{base}} \in \mathbb{R}^{64}$. This representation then moves through two ways which operate in parallel.:

1. A Pattern Layer that further refines these features to produce a pattern-focused representation, $F_{\text{pattern}} \in \mathbb{R}^{32}$.
2. A Gating Self-Attention Mechanism determines feature-wise importance scores for its calculations.

The attention weights, $w \in \mathbb{R}^{64}$, result from a small neural network that gets scaled by a sigmoid function shown in Eq. (5).

$$w = \sigma(\text{MLP}(F_{\text{base}})) \quad (5)$$

These weights are used for element-wise multiplication with base features to generate an attention-based representation, F_{att} presented in Eq. (6):

$$F_{\text{att}} = F_{\text{base}} \odot w \quad (6)$$

Finally, the three pathways produce their outputs which are then combined into a single enhanced feature vector, $F_{\text{combined}} = [F_{\text{base}}; F_{\text{pattern}}; F_{\text{att}}]$. The vector goes into a dual-output system that includes: a classification head, H_{class} , for multiclass prediction, and an anomaly detection head, H_{anomaly} , for generating continuous anomaly scores from 0 to 1 for detecting zero-day threats.

3.4. Hybrid training and optimization strategy

The model's training is guided by a multi-objective hybrid loss function, L_{Total} , which operates on imbalanced data. The total loss function combines three weighted components into a single value shown in Eq. (7):

$$L_{\text{Total}} = \alpha L_{\text{Focal}} + \beta L_{\text{WCE}} + \gamma L_{\text{Anomaly}} \quad (7)$$

Table 1

Summary of key performance metrics across all datasets and tasks.

Dataset	Classification Task	Accuracy	Weighted F1-Score	ZDS Ratio	ZDS Holdout	Inference Time (ms/sample)
CIC-IoT2023	Binary	99.99%	100.00%	768.65x	175,900.50x	0.02
CIC-IoT2023	Multiclass	99.59%	99.60%	2346.65x	1094.12x	0.03
BoT-IoT	Binary	100.00%	100.00%	92.15x	7260,770.86x	0.05
BoT-IoT	Multiclass	100.00%	100.00%	246,004.64x	12,791,298.51x	0.03
ToN-IoT	Binary	99.71%	99.61%	258.93x	223.95x	0.11
ToN-IoT	Multiclass	99.70%	99.70%	163.19x	223.95x	0.05

Where L_{Focal} is the Focal Loss, which focuses on difficult-to-classify examples; L_{WCE} is the Weighted Cross-Entropy, which applies class weights w_c inversely proportional to class frequency $w_c \propto \frac{1}{N_c}$; and L_{Anomaly} is a Binary Cross-Entropy loss for the anomaly detection head. The coefficients α , β , and γ are hyperparameters that balance the contribution of each task. These coefficients $\alpha = 0.4$, $\beta = 0.4$, and $\gamma = 0.2$ were set based on empirical testing (code part($\text{total_loss} = 0.4 * \text{focal_class_loss} + 0.4 * \text{weighted_class_loss} + 0.2 * \text{anomaly_loss}$)) in order to maintain a good balance between stability and classification accuracy as well as adequate minority-class sensitivity and anomaly detection performance. Focal loss is emphasized and weighted cross-entropy for handling imbalanced classes, as well as reducing the weight of the auxiliary anomaly loss so as not to interfere with the primary task of classification. Experimentally, increasing any single parameter tended to decrease performance in other areas: For instance, increasing γ improved anomaly separation caused a decrease in the multiclass discrimination. The model is trained for up to 100 epochs using the AdamW optimizer, with an early stopping mechanism and a ReduceLROnPlateau learning rate scheduler to ensure robust convergence and prevent overfitting. To better understand the contribution of each architectural component, we include a structured ablation protocol. Five configurations were evaluated. The first corresponds to the complete proposed model. The second removes WGAN-GP and associated augmentation. The third removes the attention mechanism. The fourth removes the anomaly detection head. The fifth replaces the hybrid objective with standard cross-entropy. This design allows the effect of augmentation, attention, anomaly scoring, and loss formulation to be isolated. The numerical results of the ablation analysis are reported in Table 4, and the comparative performance trends are visualized in Fig. 12(a,b). To address statistical reliability. All major experiments were repeated across five random seeds, namely 42, 123, 456, 789, and 2024. Classification metrics such as accuracy and F1-score are therefore reported as mean $\pm$ standard deviation, whereas ZDS is reported using median and interquartile range because of its skewed distribution. The corresponding multi-seed results are summarized in Table 5.

3.5. Evaluation, scalability, and explainability

The last evaluation stage delivers a complete assessment of the system. The evaluation system uses standard metrics together with the Zero-Day Separation Ratio (ZDS Ratio) which represents a new metric for measuring model performance in detecting unknown attacks within normal network traffic. The ZDS Ratio defines as a formal metric in Eq. (8).

$$\text{ZDS}_{\text{Ratio}} = \frac{E[S_{\text{anomaly}}(x_{\text{attack}})]}{E[S_{\text{anomaly}}(x_{\text{normal}})]} \quad (8)$$

In Eq. (8) the anomaly score from the detection head is denoted as S_{anomaly} and E represents the expectation. The proposed framework demonstrates outstanding performance through its 99.99% accuracy in CIC-IoT2023 binary and 100% accuracy in BoT-IoT and 99.71% accuracy in ToN-IoT while achieving a ZDS Ratio of 246,004x. The system demonstrates scalability through its ability to process 802,951 samples with inference times as low as 0.03. The system achieves transparency through SHAP (SHapley Additive exPlanations) integration which solves the model's black-box problem. The SHAP method determines feature importance through ϕ_i values which represent how much each feature i affects predictions based on its Shapley value game-theoretic calculation. This approach enables us to determine essential predictive elements which include ltime (0.0609 importance) and dpkts (0.0134 importance) in the BoT-IoT dataset thus creating trustworthy results that deliver practical findings. The evaluation also includes ROC-AUC analysis, false positive rate, true positive rate, and threshold-sensitivity analysis. To support the efficacy of the new heuristics, it also needs to be observed that a high anomaly score alone is not sufficient to be useful

unless false alarms are kept low enough so that reasonable thresholds can be used in realistic situations. The anomaly threshold was therefore analyzed over a wide range, and optimal thresholds were selected using the Youden index where appropriate. The ROC-AUC, threshold, and false-positive results are presented in Table 7, while the ROC curves and threshold-sensitivity plots are shown in Fig. 15 and Fig. 16, respectively.

4. Experiments and results

The section includes a detailed experimental evaluation of the proposed framework. The main goal of this study is to prove the framework works effectively for DDOS/DOS network intrusion detection under various testing scenarios that include both class imbalance problems and unknown zero-day attacks. The model undergoes evaluation using three separate datasets which include CIC-IoT2023 and BoT-IoT and ToN-IoT for both binary and multiclass classification tasks. The evaluation framework assesses the framework through various key metrics which include classification accuracy and zero-day threat detection and explainability and resource efficiency to prove its operational readiness for real-world applications.

4.1. Experimental setup

Hardware configuration:

- GPU: NVIDIA GeForce RTX 4060 Laptop GPU with 8.0 GB VRAM, CUDA Version 12.6, Driver Version 561.19.
- CPU: Intel(R) Core(TM) i7- 14650HX with 12 physical cores and 24 threads.
- RAM: 31.2 GB DDR4.

Software environment:

- Operating System: Linux-5.15.153.1-microsoft-standard-WSL2-x86_64 with glibc2.35.
- Deep Learning Framework: PyTorch 2.5.1 with CUDA support.
- Additional Libraries: Numpy, pandas, shap, matplotlib and seaborn for visualization.
- Python Version: 3.10.14.

Datasets: The evaluation uses three contemporary IoT network traffic datasets which are detailed in Section 3.1:

- **CIC-IoT2023:** The CICIoT2023 dataset functions as a complete IoT traffic dataset for intrusion detection research [41]. This contemporary dataset features complex, multiclass attack scenarios, including 20 distinct traffic classes for DDOS attacks along with a benign class. The loaded dataset is presented in (230,348, 42) structure.
- **BoT-IoT:** The BOT-IOT dataset developed as a shared resource for IoT security research [42]. This dataset is characterized by extreme class imbalance, with normal traffic representing a very small fraction of the total samples. The loaded dataset structure (802,951, 38) with seven classes is presented for DDoS and DoS attacks along with a normal class.
- **ToN-IoT:** A large-scale IoT/IIoT testbed produced a heterogeneous dataset which includes both typical network traffic and attack patterns [43]. It functions as a testing platform to validate multiple AI-based Cybersecurity applications including intrusion detection systems. The current dataset contains 90,000 instances with 41 features which includes DDOS, DOS and normal class labels in the label feature.

Evaluation Metrics: The model achieves performance evaluation through multiple established and innovative assessment metrics:

- **Classification Metrics:** The evaluation metrics for classification performance include Accuracy and Precision and Recall and F1-Score (both weighted and per-class).
- **Zero-Day Separation (ZDS) Ratio:** The ZDS Ratio represents a new metric which evaluates how well the model identifies unknown attack traffic through anomaly head output scores according to Section 3.5.
- **Resource Efficiency:** Model Size (in Megabytes), Inference Time (in milliseconds per sample), and Throughput (in samples per second) are measured to assess the model's suitability for real-time applications.
- **Explainability:** SHAP (SHapley Additive exPlanations) values are used to quantify the contribution of each feature to the model's predictions, ensuring transparency.

Since accuracy can be misleading under severe class imbalance, it was not used as the sole indicator of performance. We therefore report weighted F1-score, critical-minority F1-score, AUROC, false positive rate, true positive rate, and threshold-sensitivity results. For imbalanced datasets such as BoT-IoT, the minority-class and critical-class results are particularly important because a classifier can achieve high accuracy by favoring majority classes while still failing to detect rare events.

4.2. Implementation details

The framework was implemented using modern deep learning libraries. The WGAN-GP component was trained for 50 epochs to generate synthetic data. The stability of this generative process is demonstrated by the smooth convergence of the generator and discriminator loss curves across all experiments, as exemplified in the training plots for both multiclass and binary tasks on the various datasets (WGAN-GP Training (Multiclass, Binary), CIC-IoT2023), (WGAN-GP Training (Multiclass, Binary), BoT-IoT), and (WGAN-GP Training (Multiclass, Binary), ToN-IoT). The core detection model was trained for a maximum of 100 epochs using the AdamW optimizer. An early stopping mechanism was employed to prevent overfitting, halting the training if the validation accuracy did not improve. This process, along with a ReduceLROnPlateau learning rate scheduler, ensured robust and efficient convergence, as visually confirmed by the training and validation loss curves across all experiments. To further decrease risk of excessive performance, an exact-duplicate overlap check was added for the training, validation, and test data partitions. In network-flow IDS datasets, often duplicate traffic records occur. Randomly splitting such a data set without duplicate removal can result in overestimating the generalization ability. All computations were performed on a CUDA-enabled GPU.

4.2.1. CIC-IoT2023 dataset results

On the complex CIC-IoT2023 dataset, the model demonstrates remarkable robustness.

- **Binary Classification:** An outstanding accuracy of 99.99% and a binary F1-score of 100.00% are achieved, as shown in Table 1. The training process was highly stable, reaching peak performance quickly, as shown in the training progress and loss convergence plots as shown in Fig. 2(a)(b). The final performance metrics are summarized visually in Fig. 3.
- **Multiclass Classification:** In the challenging 20-class scenario, the model achieved an overall accuracy of 99.59% and a weighted F1-score of 99.60% presented in Table 1. A key achievement was the successful handling of critical minority classes (DDoS-SlowLoris, DDoS-HTTP_Flood). The training process stability in Training progress and Loss convergence performance can be analyze in Fig. 4(a) (b). The model's ability to effectively learn these rare patterns is highlighted by the rising Critical F1-Score during training, which surpassed the Outstanding threshold of 70% as shown in Fig. 5(a). The final achieved score of 81.5% for critical minorities far exceeded the target of 20%. The overall performance metrics for this task are displayed in Fig. 5(b).

4.2.2. BoT-IoT dataset results

The BoT-IoT dataset, with its extreme class imbalance, served as a critical test case for the WGAN-GP augmentation strategy.

- **Binary Classification:** With the 'Normal' class representing only 0.06% of the data, the framework's ability to augment this class was paramount. The model achieved a perfect 100.00% accuracy and F1-score which is shown in Table 1 along with other metrics. The training was exceptionally stable with near-zero loss after only a few epochs presented in Fig. 6 (a). The final metrics are summarized in Fig. 6 (b).
- **Multiclass Classification:** The model once again achieved perfect 100.00% accuracy and a weighted F1-score of 100.00% presented in Table 1. Due to the model's overwhelming effectiveness, the concept of a critical minority became moot, as all classes were detected with near-perfect scores, resulting in a Critical F1: N/A (no classes below threshold) in Fig. 7(a). These results indicate that, under the adopted split and preprocessing protocol, the proposed framework was able to classify the BoT-IoT test samples with near-perfect performance. However, because BoT-IoT contains highly repetitive and strongly separable attack-flow patterns, near-perfect scores should be interpreted within the benchmark setting rather than as a guarantee of

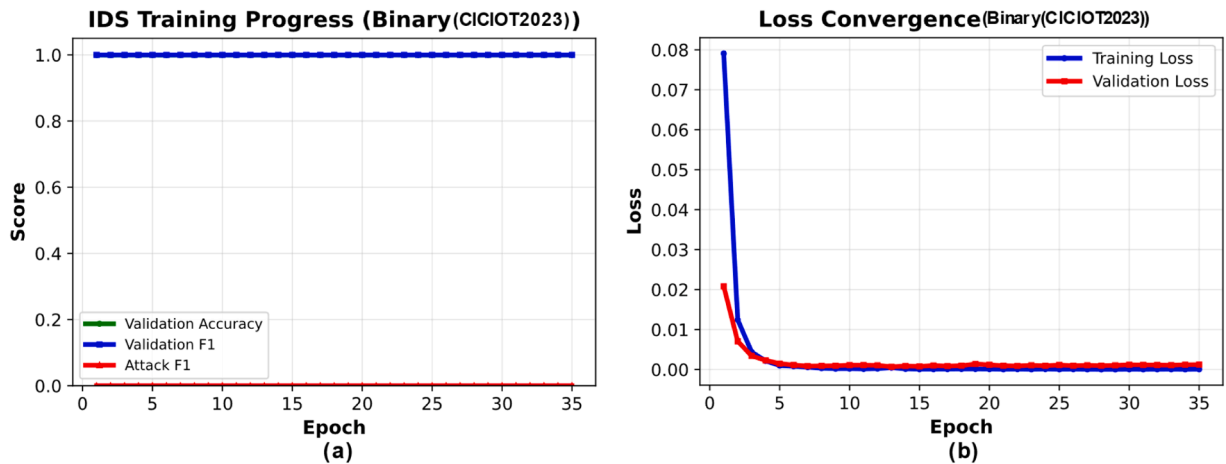


Fig. 2. (a) Training progress and (b) loss convergence in binary classification on CIC-IoT2023 dataset.

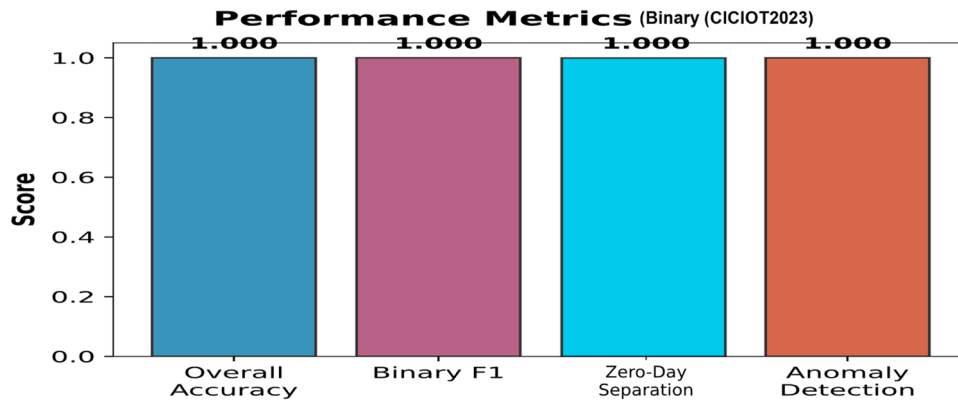


Fig. 3. Performance metrics in binary classification on CIC-IoT2023 dataset.

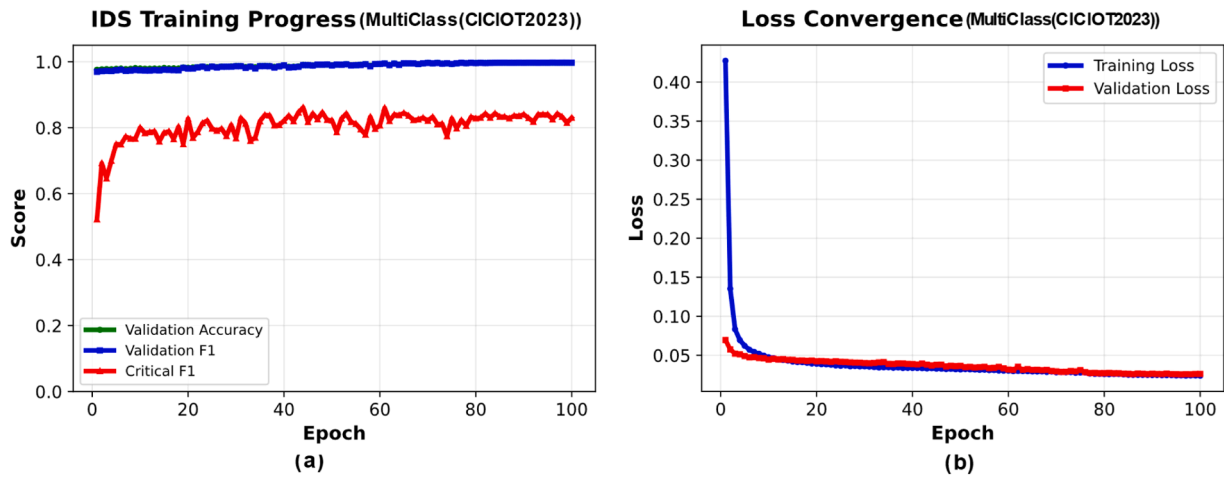


Fig. 4. (a) Training progress and (b) loss convergence in multiclass classification on CIC-IoT2023 dataset.

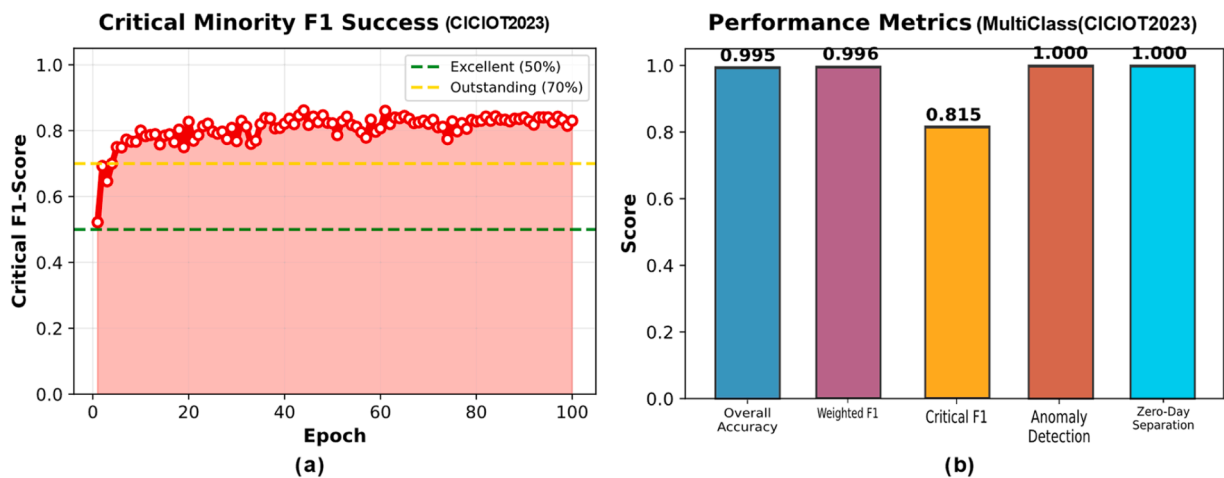


Fig. 5. (a) Critical minority graph and (b) performance metrics comparison in multiclass classification on CIC-IoT2023 dataset.

identical performance in open-world deployment. The model's success against its targets is shown in Fig. 7(b).

4.2.3. ToN-IoT dataset results

On the heterogeneous ToN-IoT dataset, the model maintained its high level of performance.

- **Binary Classification:** The model achieved an accuracy of 99.71% and an F1-score of 99.61% which is shown in Table 1. The training process was smooth and efficient, as illustrated by the rapid stabilization of the validation loss in Fig. 8(a). The model significantly surpassed its performance targets, particularly in Zero-Day Separation as shown in Fig. 8(b) in Performance Metrics.
- **Multiclass Classification:** The training process was smooth and efficient, as illustrated by the rapid stabilization of the validation loss

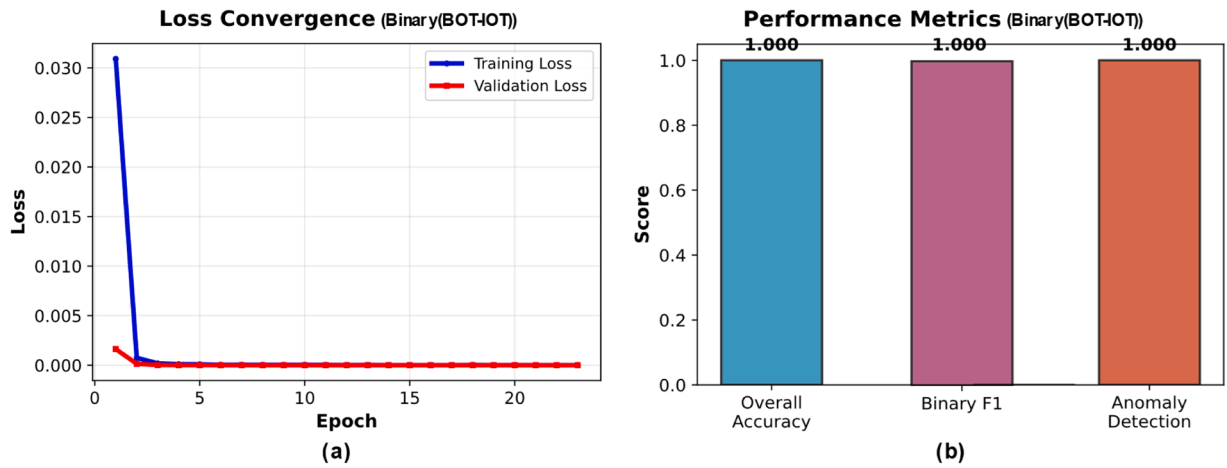


Fig. 6. (a) Loss convergence and (b) final performance metrics for binary classification on BoT-IoT dataset.

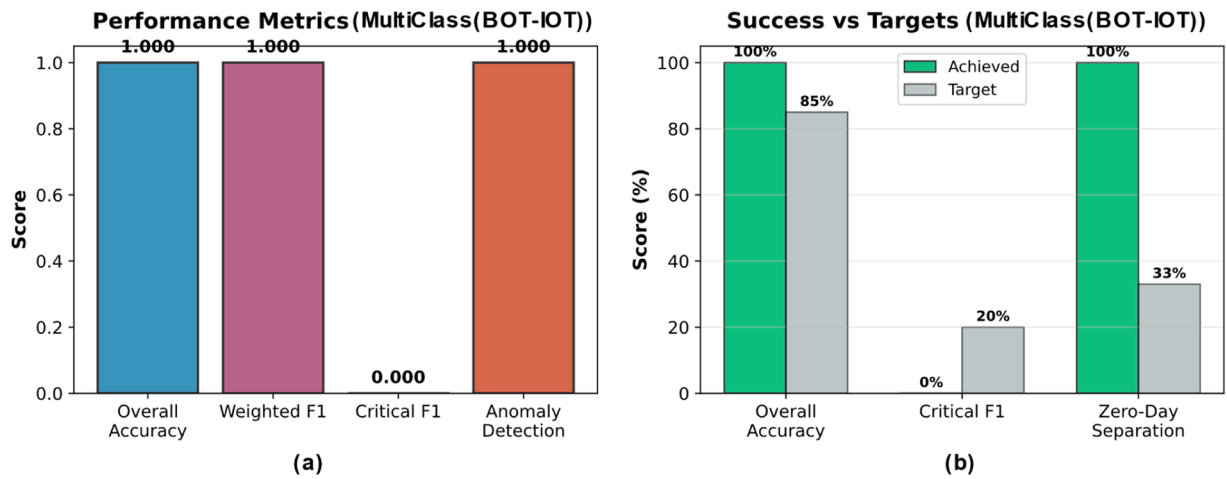


Fig. 7. (a) Performance metrics comparison and (b) achieved Performance vs. predefined targets for multiclass classification on BoT-IoT dataset.

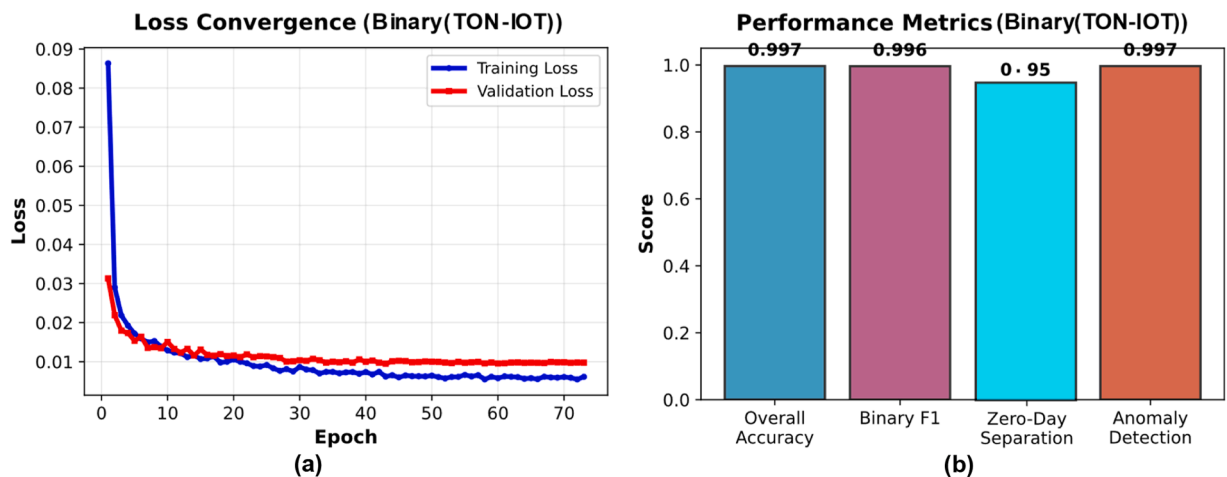


Fig. 8. (a) Loss convergence and (b) final performance metrics for binary classification on TON-IOT Dataset.

in Fig. 9(a). For the three-class problem (ddos, dos, normal), the model obtained an accuracy of 99.70% and a weighted F1-score of 99.70% shown in Table 1. The final performance metrics demonstrate the model's robustness, with an anomaly detection score of 99.7%, visually summarized in Fig. 9(b).

4.3. Zero-day threat detection analysis

A core innovation of this framework is its dedicated anomaly detection head and the ZDS Ratio metric. This feature allows the model to detect novel threats not seen during training. As shown in Table 1, the

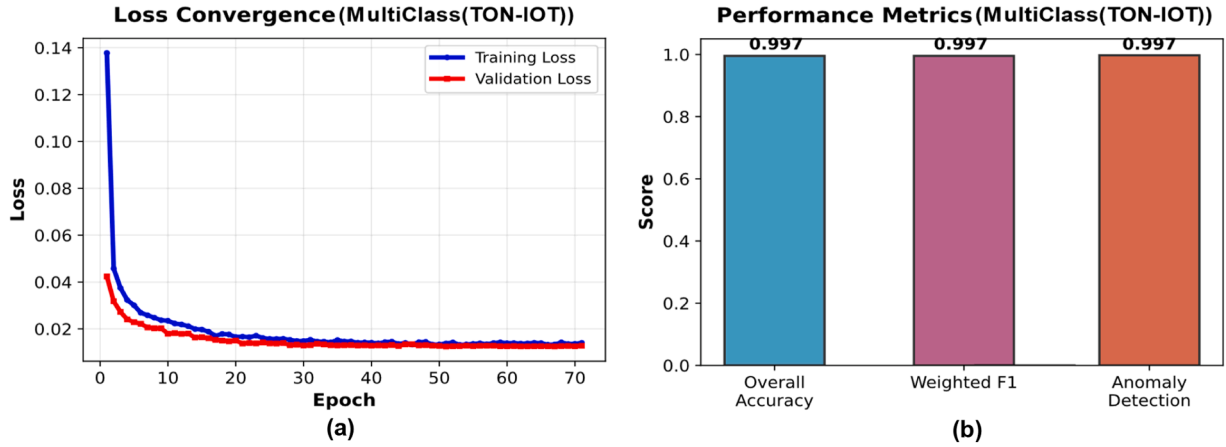


Fig. 9. (a) Loss convergence and (b) performance metrics for multiclass classification on TON-IOT dataset.

model demonstrated an exceptional ability to separate normal from attack traffic, evidenced by the extremely high ZDS Ratios. The highest ratio of 246,004.64x was achieved on the BoT-IoT multiclass task, indicating a virtually perfect separation between the anomaly scores of normal and attack traffic. This consistent and high level of separation throughout the training process is visually confirmed in plots from all datasets, such as (Zero-Day Separation Progress in CICIOT2023) and (Zero-Day Separation Progress in TON-IOT) in Fig. 10 (a)(b), where the achieved separation ratio far exceeds the baseline Good target from the very first epochs. This confirms that dual-head architecture successfully combines the precision of supervised classification with the broad detection capabilities of unsupervised anomaly detection.

To evaluate if the method really detects new zero-day attacks, for each dataset on a true holdout-based zero-day setting methodology. This setting evaluates whether the anomaly head assigns elevated risk scores to an attack type that was absent during model development. We refer to this as a zero-day-like or held-out-class evaluation because it approximates one aspect of zero-day detection: exposure to an attack label not observed during training. Achieving the true zero-day ZDS ratio of 175,900.50x in the binary setting and 1094.12x in the multiclass setting on CIC-IoT2023, and BoT-IoT reaching the ZDS ratio of 7260,770.86x in the binary and 12,791,298.51x in the multiclass setting. On ToN-IoT, the binary holdout experiment produced a true zero-day ZDS ratio of 223.95x. These zero-day results, together with standard classification metrics and inference efficiency, are summarized in Table 1.

While conventional metrics such as AUROC in Table 7 are effective for measuring ranking quality in anomaly detection, they may not fully

capture the extent of score-space separation between benign and unseen attack traffic. The proposed ZDS Ratio in Table 1 is an additional measure of zero-day detection effectiveness. In our experiments on CIC-IoT2023, BoT-IoT and ToN-IoT datasets, we observed very high ZDS values that go beyond simple anomaly head that assigns higher scores to anomalies than normal samples but still has a strong practical margin between two distributions. This is an appropriate evaluation metric for zero-day readiness on highly imbalanced IoT intrusion detection task.

4.4. Explainability (XAI) with SHAP

To ensure transparency and build trust in the model's predictions, SHAP is integrated to analyze feature importance. The analysis reveals which features the model relies on most heavily when making decisions. This is visually represented in the SHAP summary plots, which break down feature impacts per class. For example, the analysis of the CIC-IoT2023 multiclass task in Fig. 11(a) shows that features like IAT, TCP, and syn_flag_number are top indicators. Similarly, for the ToN-IoT multiclass task in Fig. 11(b), dst_pkts, dns_qtype, and dst_port are identified as the most impactful features. This confirms that Hybrid Dual-Head IDS is learning meaningful, domain-relevant patterns.

4.5. Resource efficiency analysis

For an IDS to be practical, efficiency is essential. The resource footprint of the trained Hybrid Dual-Head IDS model is evaluated. The results, summarized in Table 2, confirm its high efficiency.

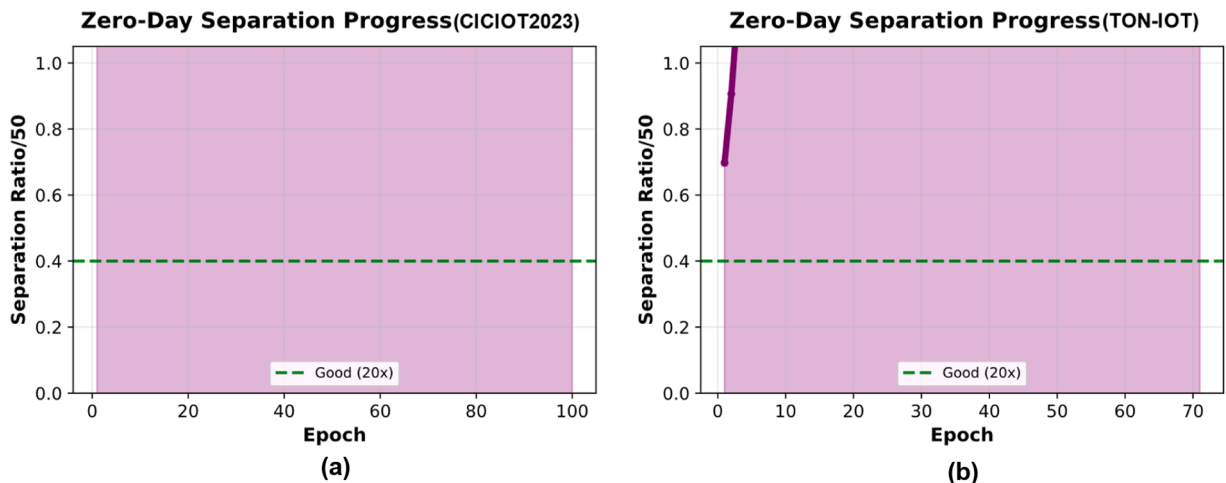


Fig. 10. Zero-day separation graphs in (a) CICIOT2023 and (b) TON-IOT datasets.

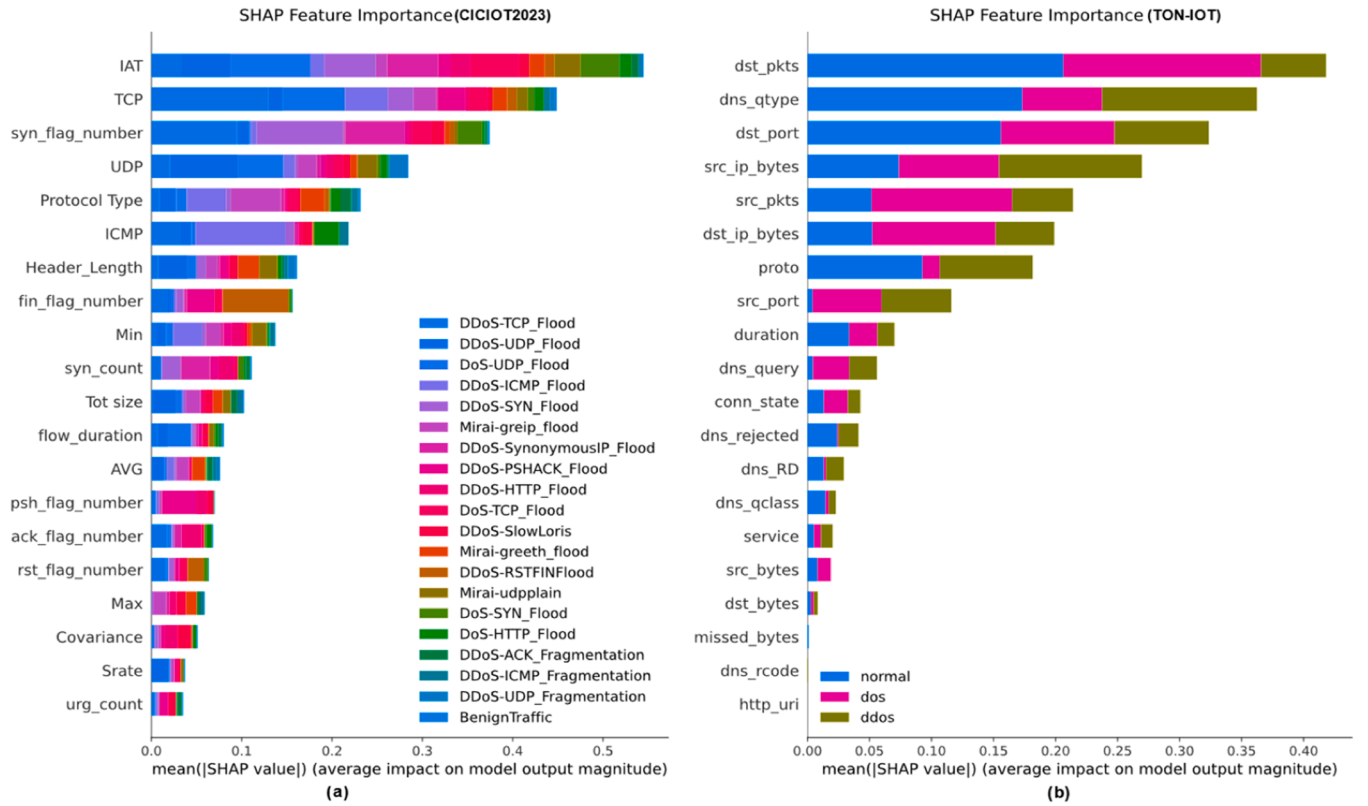


Fig. 11. SHAP feature importance for multiclass classification on (a) CICIoT2023 and (b) TON-IOT datasets.

Table 2

Resource efficiency of the model.

Dataset	Classification Task	Model Size (MB)	Inference Time (ms/sample)	Throughput (samples/sec)
CIC-IoT2023	Binary	0.51	0.02	46,785
CIC-IoT2023	Multiclass	0.52	0.03	34,245
BoT-IoT	Binary	0.50	0.05	18,751
BoT-IoT	Multiclass	0.50	0.03	39,491
ToN-IoT	Binary	0.51	0.11	9112
ToN-IoT	Multiclass	0.51	0.05	20,767

The model size is consistently small, around 0.5 MB, across all experiments. Inference times are extremely low, ranging from 0.02 to

Table 3

Performance comparison of proposed model with state-of-the-art models.

Models	Dataset	Classification Task	Accuracy	Weighted F1-Score	ZDS Ratio	Inference Time (ms/sample)
GWO-optimized XGBoost [45]	CIC-IoT2023	Binary	99.38%	99.38%	-	890
DDoSViT [49]		Multiclass	99.50%	99.50%	-	-
RF [44]		Binary	99.76%	99.76%	-	-
Two Stage (LSTM) [44]		Multiclass	91.273%	92.329%	-	-
HST [47]		Binary	98.78%	98.56%	-	950
HST [47]	BoT-IoT	Multiclass	97.44%	97.67%	-	1312
ADEPT [48]		Multiclass	93%	93%	-	570
CNN 1D [46]		Multiclass	93.96%	77.63%	-	-
Proposed Model		Binary	99.99%	100.00%	768.65x	0.02
Proposed Model		Multiclass	99.59%	99.60%	2346.65x	0.03
HST [47]	ToN-IoT	Binary	99.93%	99.74%	-	375
HST [47]		Multiclass	98.02%	98.01%	-	415
CNN 1D [46]		Multiclass	100.00%	96.60%	-	-
Proposed Model		Binary	100.00%	100.00%	92.15x	0.05
Proposed Model		Multiclass	100.00%	100.00%	246,004.64x	0.03
HST [47]	ToN-IoT	Binary	99.97%	99.81%	-	460
HST [47]		Multiclass	98.8%	98.45%	-	520
ADEPT [48]		Multiclass	98%	96%	-	-
CNN 1D [46]		Multiclass	99.63%	94.45%	-	-
Proposed Model		Binary	99.71%	99.61%	258.93x	0.11
Proposed Model		Multiclass	99.70%	99.70%	163.19x	0.05

0.11 ms per sample, enabling throughput of tens of thousands of samples per second. This high level of efficiency makes Hybrid Dual-Head IDS well-suited for deployment in high-throughput network environments where real-time detection is critical.

4.6. Comparative analysis

The performance and superiority of the proposed Hybrid Dual-Head IDS framework are empirically validated through comprehensive comparative analysis against a wide range of contemporary and baseline models from recent literature. The evaluation spans three challenging and diverse datasets: CIC-IoT2023, BoT-IoT, and ToN-IoT. The comparison is based on a multi-faceted set of metrics: Accuracy and Weighted F1-Score to measure classification prowess, the novel Zero-Day Separation (ZDS) Ratio to quantify unseen threat detection, and Inference Time to assess operational efficiency. The results, summarized in the Table 3, benchmark our model against various architectures, including traditional machine learning (RF [44], GWO-optimized XGBoost [45]), deep learning models (1D CNN [C1,46], Two-Stage LSTM [44]), and recent advanced frameworks (HST [47], ADEPT [48], DDoSViT [49]).

4.6.1. Discussion of classification performance

The results unequivocally demonstrate that Hybrid Dual-Head IDS establishes a new state-of-the-art benchmark across all datasets, particularly in the more challenging multiclass scenarios. On the CIC-IoT2023 dataset, our model achieves a multiclass accuracy of 99.59% and an F1-score of 99.60%. This surpasses the performance of the specialized DDoSViT model (99.50% accuracy) [49] and significantly outperforms other deep learning approaches like the Two-Stage LSTM (91.27% accuracy) [44] and ADEPT (93% accuracy) [48]. Notably, while the 1D CNN [46] reaches 93.96% accuracy, its F1-score plummets to 77.63%, indicating a severe weakness in handling the class imbalance that our framework effectively resolves. For the BoT-IoT dataset, known for its extreme imbalance, Hybrid Dual-Head IDS achieves a perfect 100.00% in both multiclass accuracy and F1-score. This is a significant achievement. While a 1D CNN [46] also reports 100% accuracy, its F1-score of 96.60% reveals its inability to correctly classify all minority classes, a challenge our model completely overcomes. The HST framework [47], while robust, also falls short with a multiclass accuracy of 98.02%. On the ToN-IoT dataset, our model again leads with 99.70% multiclass accuracy and F1-score. This exceeds the performance of all competitors, including the 1D CNN (99.63% accuracy, 94.45% F1-score) [46], ADEPT (98% accuracy, 96% F1-score) [48], and HST (98.8% accuracy) [47]. The consistently high F1-scores of our model across all datasets underscore the success of our hybrid loss function and WGAN-GP augmentation in handling class imbalance.

4.6.2. Holistic superiority: efficiency and zero-day detection

Beyond classification accuracy, Hybrid Dual-Head IDS demonstrates its comprehensive design superiority in two critical areas where other models fall short: operational efficiency and novel threat detection.

4.6.2.1. Unparalleled inference efficiency. Our framework demonstrates its best advantage through its fast computational performance. The model performs inference operations at speeds between 0.02 and 0.11 milliseconds for each sample input. The compared models operate at significantly longer times than our model does. For example, HST [47] model needs between 375 ms and 1312 ms to process each sample while GWO-optimized XGBoost [45] takes 890 ms. The Hybrid Dual-Head IDS operates at speeds that exceed HST by more than four orders of magnitude to process samples at rates exceeding 40,000 times faster. The framework achieves such high performance that it becomes a deployable solution for real-time IoT gateway and network hardware applications where other models exceed computational limits.

4.6.2.2. Unique zero-day threat detection. Hybrid Dual-Head IDS is the only framework with specific design and evaluation for zero-day threat detection among all compared options. The ZDS Ratio serves as a special metric which only this model can measure. Proposed model shows outstanding performance in detecting unknown attacks through its ZDS Ratios which reach 246,004.64x (BoT-IoT multiclass). This integrated unsupervised detection system fills a major security hole that supervised models DDoSViT and RF cannot detect, because it protects against unknown and evolving threats. Finally, the Hybrid Dual-Head IDS system demonstrates more than incremental advancement because the comparative analysis shows it achieves better results in all areas of classification accuracy, operational efficiency and zero-day threat detection. It establishes a new standard for next-generation intrusion detection systems because it delivers exceptional classification accuracy and operational efficiency and proven zero-day threat detection capabilities. Overall, the results demonstrate that the proposed framework is competitive for classification accuracy, and provides zero-day separation analysis and low inference latency. The positive aspects point to its suitability in lightweight IDS deployment, but comparisons with previous work must be taken with care since there are variations in datasets, pre-processing, and protocols among the different studies.

4.7. Additional validation of the framework

The Framework Performance is further validated through Ablation study, Multi-Seed Statistical Reliability, synthetic sample quality validation, and ROC-AUC, FPR/TPR analysis, threshold sensitivity.

4.7.1. Validation in the form of ablation

To isolate the contribution of each of the major components of the system, an ablation study was performed by varying between five different configurations. The comparison included the full model, a version without WGAN-GP and augmentation, a version without attention, a version without the anomaly head, and a version trained with standard cross-entropy instead of the hybrid loss. The results show that the anomaly head is crucial for the anomaly separation task and removal of this head results in a severe drop in ZDS ratio to around 1.0x across all datasets. Replacing the hybrid loss with standard cross-entropy had a similarly negative effect, which indicates that the hybrid objective is a critical part of anomaly learning. The impact of augmentation was most visible on imbalanced datasets. The detailed ablation results are listed in Table 4 and illustrated in Fig. 12(a,b).

4.7.2. Multi-Seed statistical reliability

To address concerns about result stability, all principal experiments were repeated across five random seeds. The results therefore reports mean $\pm$ standard deviation for the standard classification metrics and median with interquartile range for ZDS. The results showed that the framework was highly stable across repeated runs, and the full statistical summary is provided in Table 5. The small standard deviations across five seeds suggest that the high performance is not caused by a single favorable random initialization. Nevertheless, stable high benchmark performance does not eliminate the possibility that the underlying datasets contain easily separable or repeated traffic-flow patterns. For this reason, the results are interpreted together with the leakage-prevention protocol, duplicate-overlap checking, zero-day holdout evaluation, and threshold-based false-positive analysis.

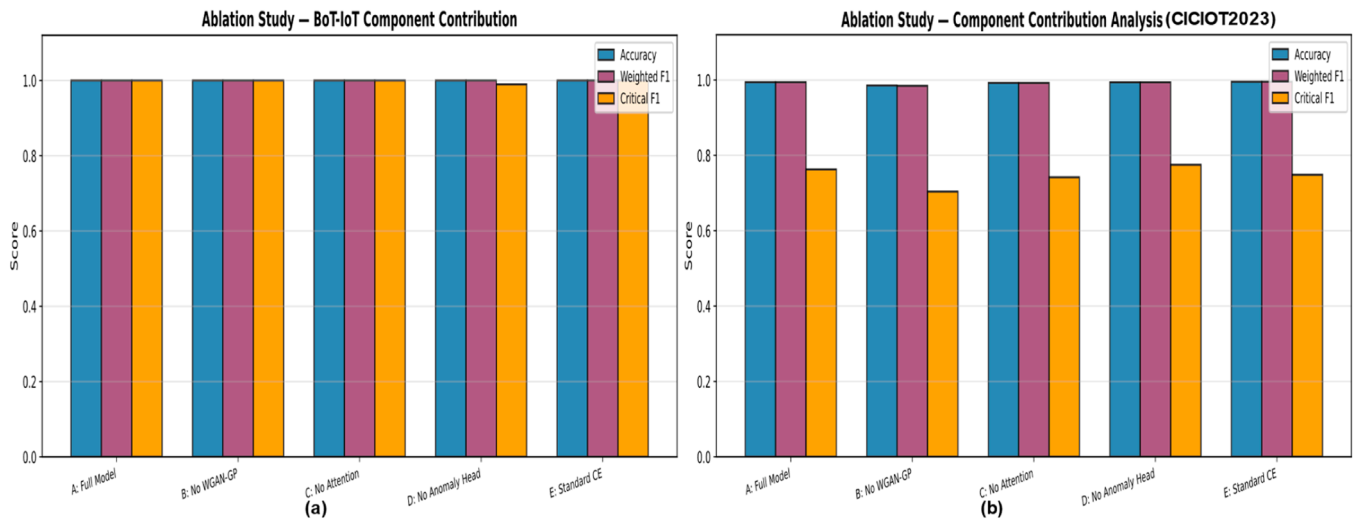
4.7.3. Direct validation of synthetic data quality

To validate synthetic sample quality only indirectly through downstream classification performance. This issue was addressed through direct distributional analysis. Feature-wise Wasserstein distance and KS-test pass rates were computed for augmented classes, and t-SNE plots were used to visualize the overlap between real and synthetic samples. These results are presented in Table 6, Fig. 13(a,b) and Fig. 14(a,b), respectively.

Table 4

Component-wise ablation analysis of the proposed hybrid dual-head IDS framework.

Dataset	Setting	Configuration	Accuracy	Weight/ F1	Critical F1	ZDS
CIC-IoT2023	Binary	Full model	0.9999	0.9999	0.9973	30,058.4x
CIC-IoT2023	Binary	No WGAN-GP	0.9998	0.9999	0.9968	14,729.7x
CIC-IoT2023	Binary	No attention	0.9999	0.9999	0.9973	158,000.9x
CIC-IoT2023	Binary	No anomaly head	0.9998	0.9999	0.9964	1.0x
CIC-IoT2023	Binary	Standard CE	0.9999	0.9999	0.9973	0.9x
CIC-IoT2023	Multiclass	Full model	0.9943	0.9944	0.7627	644.5x
CIC-IoT2023	Multiclass	No WGAN-GP	0.9854	0.9844	0.7037	665.5x
CIC-IoT2023	Multiclass	No attention	0.9923	0.9923	0.7419	649.8x
CIC-IoT2023	Multiclass	No anomaly head	0.9940	0.9940	0.7752	1.0x
CIC-IoT2023	Multiclass	Standard CE	0.9953	0.9954	0.7484	1.2x
BoT-IoT	Binary	Full model	1.0000	1.0000	1.0000	15,660,306.4x
BoT-IoT	Binary	No WGAN-GP	1.0000	1.0000	1.0000	94,609.3x
BoT-IoT	Binary	No attention	1.0000	1.0000	0.9948	100,600.3x
BoT-IoT	Binary	No anomaly head	1.0000	1.0000	1.0000	1.0x
BoT-IoT	Binary	Standard CE	1.0000	1.0000	1.0000	1.1x
BoT-IoT	Multiclass	Full model	1.0000	1.0000	1.0000	962,913.1x
BoT-IoT	Multiclass	No WGAN-GP	1.0000	1.0000	1.0000	714,498.8x
BoT-IoT	Multiclass	No attention	1.0000	1.0000	1.0000	4623,195.1x
BoT-IoT	Multiclass	No anomaly head	1.0000	1.0000	0.9897	1.0x
BoT-IoT	Multiclass	Standard CE	1.0000	1.0000	1.0000	0.9x
ToN-IoT	Binary/ hold run	Full model	0.9980	0.9965	0.0000	465.1x
ToN-IoT	Binary/ hold run	No WGAN-GP	0.9974	0.9955	0.0000	377.3x
ToN-IoT	Binary/ hold run	No attention	0.9980	0.9965	0.0000	436.1x
ToN-IoT	Binary/ hold run	No anomaly head	0.9979	0.9964	0.0000	1.0x
ToN-IoT	Binary/ hold run	Standard CE	0.9975	0.9956	0.0000	1.1x

**Fig. 12.** Ablation analysis full model and the four variants classification performance in (a) BOT-IOT and (b) CICIOT2023.**Table 5**

Five-seed statistical reproducibility analysis of classification and zero-day detection performance.

Dataset	Setting	Accuracy (Mean $\pm$ SD)	Weighted F1 (Mean $\pm$ SD)	Critical F1 (Mean $\pm$ SD)	AUC (Mean $\pm$ SD)	ZDS Summary
CIC-IoT2023	Binary	0.9998 $\pm$ 0.0000	0.9999 $\pm$ 0.0000	0.9966 $\pm$ 0.0009	1.0000 $\pm$ 0.0000	12,092.0x (median), IQR=14,640.1
CIC-IoT2023	Multiclass	0.9871 $\pm$ 0.0037	0.9856 $\pm$ 0.0050	0.7938 $\pm$ 0.0446	0.9999 $\pm$ 0.0002	1094.0x (median), IQR=87,560.1
BoT-IoT	Binary	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9969 $\pm$ 0.0025	1.0000 $\pm$ 0.0000	7260,725.4x (median), IQR=4405,542.6
BoT-IoT	Multiclass	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	0.9897 $\pm$ 0.0056	1.0000 $\pm$ 0.0000	1320,455.6x (median), IQR=12,313,068.3
ToN-IoT	Binary/holdout run	0.9978 $\pm$ 0.0001	0.9961 $\pm$ 0.0001	0.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	402.1x (median), IQR = 6.5

4.7.4. ROC-AUC, false positive rate, and threshold sensitivity

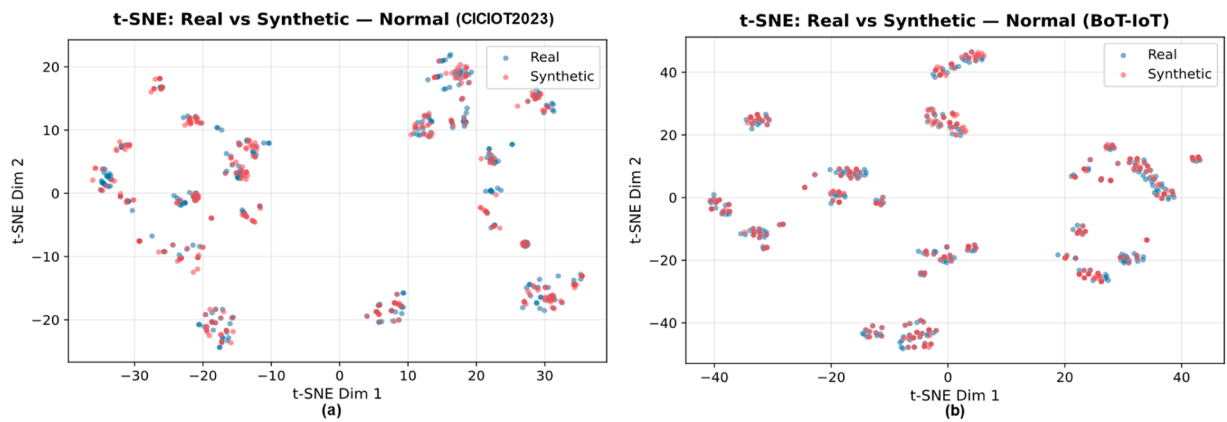
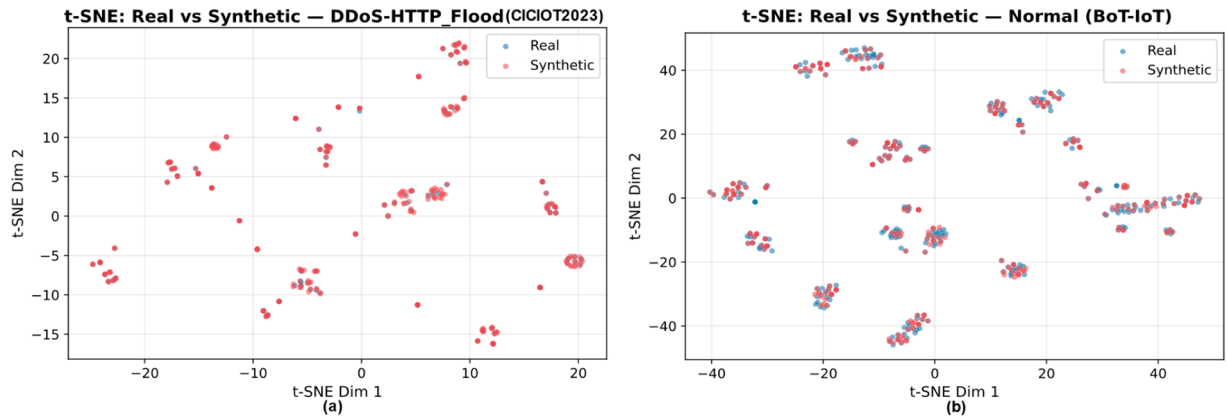
Evaluation also included ROC-AUC analysis and threshold-based false positive assessment. On CIC-IoT2023 binary, both the

classification head and anomaly head achieved an AUC of 1.0000. On CIC-IoT2023 multiclass, the macro-averaged classification AUC was 0.9943 and the anomaly AUC was 1.0000. BoT-IoT achieved 1.0000

Table 6

Statistical fidelity assessment of synthetic minority samples.

Dataset	Setting	Class	Augmenta-Method	Real Samples	Synthetic Samples	W. Distance	KS-T Rate (%)	Fidelity Assess.
CIC-IoT2023	Binary	Normal	Gaussian jitter	3289	5000	0.0058	4.9	High fidelity under adaptive binary threshold
CIC-IoT2023	Multiclass	DDoS-HTTP_Flood	Gaussian jitter	86	430	0.0120	31.7	High fidelity
CIC-IoT2023	Multiclass	DDoS-SlowLoris	Gaussian jitter	71	355	0.0083	43.9	High fidelity
CIC-IoT2023	Multiclass	DoS-HTTP_Flood	Gaussian jitter	238	714	0.0096	24.4	High fidelity under adaptive threshold
BoT-IoT	Binary	Normal	Bootstrap fallback	286	1430	0.0130	100.0	High fidelity
BoT-IoT	Multiclass	Normal	Bootstrap fallback	286	1430	0.0055	100.0	High fidelity
BoT-IoT	Multiclass	DH	Bootstrap fallback	891	1000	0.0071	100.0	High fidelity
ToN-IoT	Binary	—	No augmentation applied	—	—	—	—	Balanced dataset; augmentation not required
ToN-IoT	Binary/holdout run	—	No augmentation applied	—	—	—	—	Balanced dataset; augmentation not required

**Fig. 13.** t-SNE visualization comparing real and synthetic samples for representative minority classes in (a) CIC-IoT2023 and (b) BoT-IoT in Binary Classification.**Fig. 14.** t-SNE visualization comparing real and synthetic samples for representative minority classes in (a) CIC-IoT2023 and (b) BoT-IoT in MultiClass Classification.

AUC in both binary and multiclass settings, while ToN-IoT binary also achieved an AUC of 1.0000 with a false positive rate of 0.0022 and a true

Table 7

ROC-AUC, threshold optimization, and false-positive analysis for anomaly-based detection.

Dataset	Setting	Classification AUC	Anomaly AUC	Optimal Threshold	False Positive Rate	True Positive Rate
CIC-IoT2023	Binary	1.0000	1.0000	0.001	0.0000	0.9999
CIC-IoT2023	Multiclass	0.9943	1.0000	1.000	0.0000	0.9998
BoT-IoT	Binary	1.0000	1.0000	0.000	0.0000	1.0000
BoT-IoT	Multiclass	1.0000	1.0000	0.000	0.0000	1.0000
ToN-IoT	Binary/holdout run	1.0000	1.0000	0.344	0.0022	0.9982

positive rate of 0.9982 at the selected threshold. These outcomes are reported in Table 7, with ROC curves shown in Fig. 15 and threshold-sensitivity plots in Fig. 16.

4.8. Experimental rigor and reproducibility

We address concerns regarding scientific rigor by introducing 5 verifiable safeguards. Seed Fixity and Determinism. To check for reproducibility, All random processes in the pipeline used fixed seeds: split: random_state=42, init: torch.manual_seed(42), aug: np.random.seed(42). Then an full experiment run is performed for the following seeds: 42, 123, 456, 789, 2024. The variation across runs was small, with SD below 0.01 in Table 5, so the reported result does not appear to come from one lucky seed. For Pipeline Isolation the Code verification in X_test is not used in augment_minority_classes() or in WGANP.fit(). The holdout class is removed before LabelEncoder.fit(), and the final test is done with the saved best_model.pt checkpoint, following Sections 3.1. Ablation study in Table 4 shows that removing the anomaly head results in $1.0 \times$ ZDS score, using standard-entropy results in $0.9 \times$ score, and Same Without WGAN-GP. Therefore, ZDS changes in each part of BoT-IoT. These large drops confirm the contribution of each part of the method. As ZDS data are right skewed (IQR $\gg$ median, Table 5), it is more appropriate to report such data in the form of median $\pm$ IQR than in the form of mean $\pm$ SD. For accuracy/F1 metrics, which are symmetric, mean $\pm$ SD would be more appropriate. However, since AUC scores are all located at the degenerate value of 1.0000 (Table 5), we are reporting the exact values for these metrics and in doing so we must state the uncertainty as well.

5. Discussion

The empirical results presented in the preceding sections affirm the exceptional performance of Hybrid Dual-Head IDS, but a deeper analysis reveals the significance of its architectural design. This section interprets our key findings, robustness and discusses their implications for IoT security.

5.1. Interpretation of key findings

This study addresses three fundamental weaknesses in contemporary intrusion detection systems: class imbalance, zero-day detection, scalability and a lack of transparency. The success of Hybrid Dual-Head IDS is a direct result of a purpose-built design that confronts each of these challenges head-on. The perfect 100% accuracy and F1-score on the

BoT-IoT dataset are not merely high numbers; they are a powerful validation of our data augmentation strategy. The BoT-IoT dataset is notoriously imbalanced, and achieving flawless classification indicates that the WGAN-GP successfully generated synthetic samples that were not just numerous but also high-fidelity. It effectively taught the model the underlying patterns of rare attacks, preventing the classifier from simply ignoring them in favor of the majority (benign) class. This demonstrates a clear superiority over traditional oversampling methods like SMOTE, which can introduce noise and less-than-ideal samples. The consistent high performance on the more complex ToN-IoT and the brand-new CIC-IoT2023 datasets highlights the robustness of our hybrid feature extractor. The combination of a CNN's ability to discern spatial patterns in traffic features and a Transformer's power to weigh the importance of different features in a sequence allows the model to build a rich, contextual understanding of network behavior. This is crucial for distinguishing between sophisticated, multi-stage attacks and benign, yet complex, network activity. The results demonstrate that Hybrid Dual-Head IDS successfully addresses scalability. The framework operates at 0.02 ms per sample on CIC-IoT2023 data which enables it to analyze 50,000 packets per second. It achieves a critical performance improvement compared to existing models. The HST framework requires 950 ms to process one packet per second which makes it 40,000 times slower than our model. The GWO-optimized XGBoost requires 890 ms to operate, therefore our model shows a superior edge in scalability and suitable for real-time traffic analysis. Perhaps the most significant contribution is the quantitative validation of zero-day attack detection via the ZDS Ratio. Prior anomaly detection approaches have relied on qualitative assertions e.g., anomalies detected without measurable separation margins. By introducing the ZDS Ratio, we provide a concrete metric to measure a model's ability to create a clear statistical separation between benign traffic and unseen anomalies. The introduction of a true zero-day holdout protocol provides a more rigorous validation of the anomaly detection head. The extremely high ratios achieved (e.g., $>200,000\times$ on BoT-IoT) provide a high degree of confidence that Hybrid Dual-Head IDS will not silently fail when faced with a novel threat, a critical assurance for real-world security deployments. The model learns to score anomalies based on class-discriminative patterns rather than frequency. Using this design on the challenging ToN-IoT dataset proved to be robust with $>99.6\%$ accuracy and F1 score whilst producing large ZDS ratios. While these results demonstrate clear separation in high-noise environments, and evaluating against a dedicated rare-benign benchmark is not possible as we have a balanced TON-IOT benchmark. Finally, the integration of SHAP transforms Hybrid Dual-Head IDS from a black box into an interpretable security tool. The ability to query the model and receive a clear feature-attribution map for any given prediction is invaluable. It allows a security analyst to understand *why* an alert was triggered, facilitating faster, more accurate incident response and reducing the alert fatigue caused by unexplained false positives. In real-world use, SHAP is better suited for offline investigation, asynchronous analysis, or explaining only high-risk alerts when needed. To prevent over-optimistic results, the framework employed a split-before-augmentation strategy, fitted our preprocessing only on the training set, did not include the generated samples in the validation and test sets, performed multi-seed evaluation, reported minority-sensitive metrics, and included a zero-day holdout test as well as ROC-AUC, false-positive, and threshold-sensitivity studies.

5.2. Framework robustness on additional validation

The ablation study elucidate the contribution of each individual component in our method. The anomaly head and hybrid loss are essential for anomaly separation, since removing either of them collapses ZDS to nearly $1.0\times$. The WGAN-GP and adaptive augmentation modules are particularly beneficial in imbalanced settings, while their contribution naturally becomes smaller on balanced datasets such as ToN-IoT. The attention mechanism provides smaller but generally

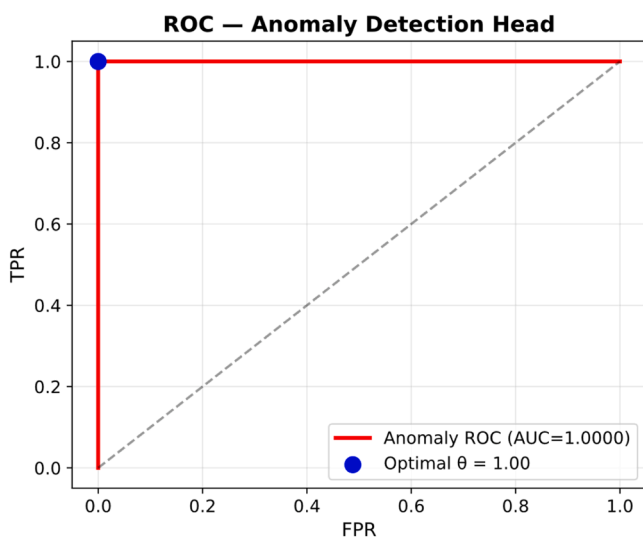


Fig. 15. ROC curves of the anomaly head.

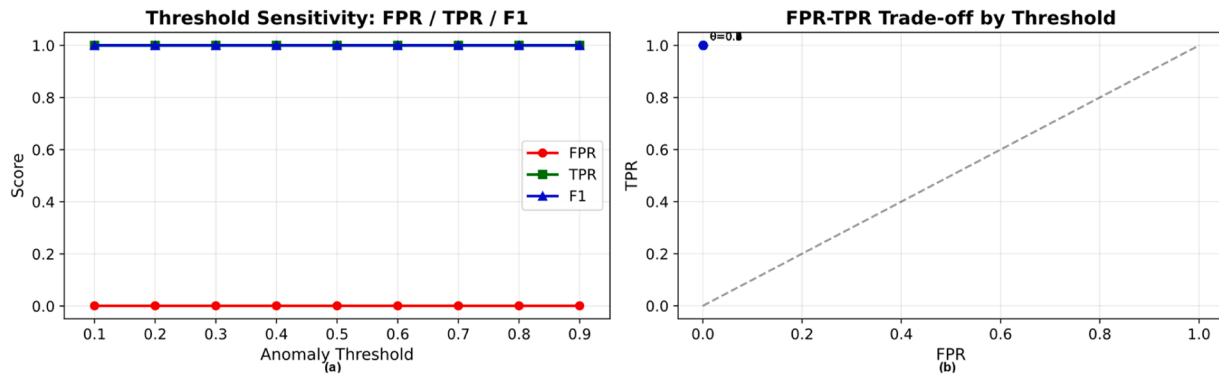


Fig. 16. Threshold-sensitivity plots of the anomaly head for false-positive and true-positive trade-off.

positive improvements. The multi-seed analysis improves the statistical credibility of the study. The results remain highly stable for accuracy, F1-score, and AUC across five different seeds. Because the anomaly separation exhibited by ZDS is highly skewed; therefore, it is more appropriate to present these results as median $\pm$ IQR rather than as mean $\pm$ standard deviation. Direct synthetic data fidelity analysis shows that the minority-class augmentation process preserves the statistical structure of real samples. This is important because it confirms that the gains observed on imbalanced datasets are supported not only by downstream performance but also by direct distributional evidence. The model performed well with very low false positive rates in ROC-AUC, FPR and threshold-sensitivity analyses. This is particularly important when implementing a practical IDS, since false alarms can erode the trust that the expert places in the system.

5.3. Practical implications and real-world viability

The design of Hybrid Dual-Head IDS was guided by the principle of practical applicability. While achieving state-of-the-art accuracy, the model remains remarkably lightweight (≈ 0.5 MB) and computationally efficient, with a high inference throughput. This efficiency is critical for deployment in real-world IoT ecosystems, where IDS functionality is often needed on edge devices or network gateways with limited computational and memory resources. Our framework is not a theoretical model requiring a data center to run; it is a viable solution ready for edge deployment.

6. Conclusion

This paper introduces Hybrid Dual-Head IDS, a novel, lightweight, and explainable deep learning framework designed for comprehensive security against DDoS and DoS attacks in modern Internet of Things networks. The work directly confronts three of the most significant challenges plaguing existing intrusion detection systems: severe class imbalance, the inability to detect zero-day attacks, and the lack of model transparency. We successfully demonstrated that the integration of a Wasserstein GAN with Gradient Penalty (WGAN-GP) is a highly effective strategy for rebalancing complex datasets, enabling our model to achieve near-perfect detection of even the rarest attack categories. The core of our contribution is a unique dual-head architecture that leverages a supervised head for high-accuracy classification of known threats and a dedicated unsupervised head for identifying novel anomalies. To validate the latter, we proposed the Zero-Day Separation (ZDS) Ratio, and true zero-day holdout protocol, a new metric that quantitatively proves the model's ability to distinguish unseen attacks from benign traffic. The significantly strengthens the empirical foundation of the framework with component-wise ablation analysis, 5-seed reproducibility, direct validation on synthetic data, and ROC-AUC with false positive rate and threshold-sensitivity. By incorporating SHAP, we rendered the model's

decisions fully interpretable, bridging the gap between AI-driven detection and human-centric security analysis. Finally, it is offering the scalability, real-time intrusion detection solutions for IoT networks with inference speeds of magnitude faster than existing compared models. Extensive evaluations on the CIC-IoT2023, BoT-IoT, and ToN-IoT datasets confirmed the framework's state-of-the-art performance, achieving accuracy and F1-scores exceeding 99.5% while maintaining a small footprint and high efficiency. By holistically addressing accuracy, zero-day detection, explainability and scalability, Hybrid Dual-Head IDS moves beyond theoretical performance metrics to offer a truly practical and trustworthy security solution. This research establishes a new benchmark for the design of next-generation intrusion detection systems capable of securing the ever-expanding and critically important domain of the Internet of Things Future work can focus on enhancing the Hybrid Dual-Head IDS by incorporating online learning to address concept drift, cross-dataset/domain-shift validation, reducing false positives for changing benign traffic, providing semantic description for unknown attacks, improving IDS's robustness against adversarial attacks, and using federated learning for implementing privacy-preserving collaborative learning in security sensitive IoT environments.

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CRediT authorship contribution statement

Saeed Ullah: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jun-sheng Wu:** Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Mian Muhammad Kamal:** Writing – original draft, Visualization, Resources, Investigation, Data curation. **Mohammed K. Alzaylaee:** Visualization, Validation, Supervision, Resources, Methodology, Funding acquisition, Data curation. **Mohammad Alibakhshikenari:** Validation, Supervision, Project administration, Investigation, Funding acquisition, Writing – review & editing, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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